

When Is It Worth Training Your Own Model?

Yoruba from scratch against multilingual transfer

A2-NLP Project Team · CSED 504 · University of Washington · Summer 2026

TAKEAWAY

A 33.8M-parameter Yoruba model is ahead of mmBERT on topic classification; the strongest evidence points to vocabulary fit—not text scarcity—as the leverage.

RESEARCH QUESTION

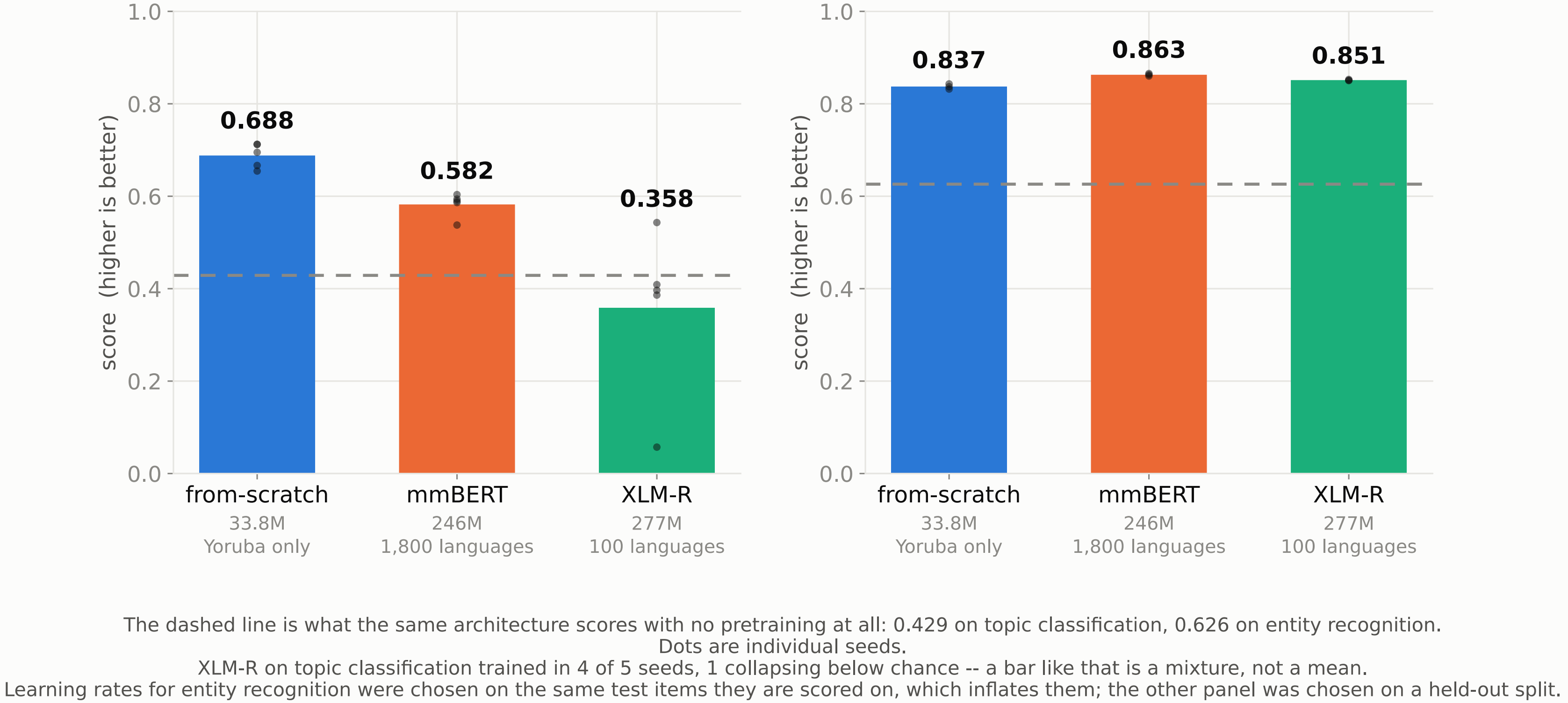
The choice

Low-resource NLP usually defaults to transfer: start from a multilingual encoder and fine-tune it. We tested the alternative—pretrain a small encoder only on Yoruba, with a vocabulary fitted to Yoruba—against XLM-R and mmBERT.

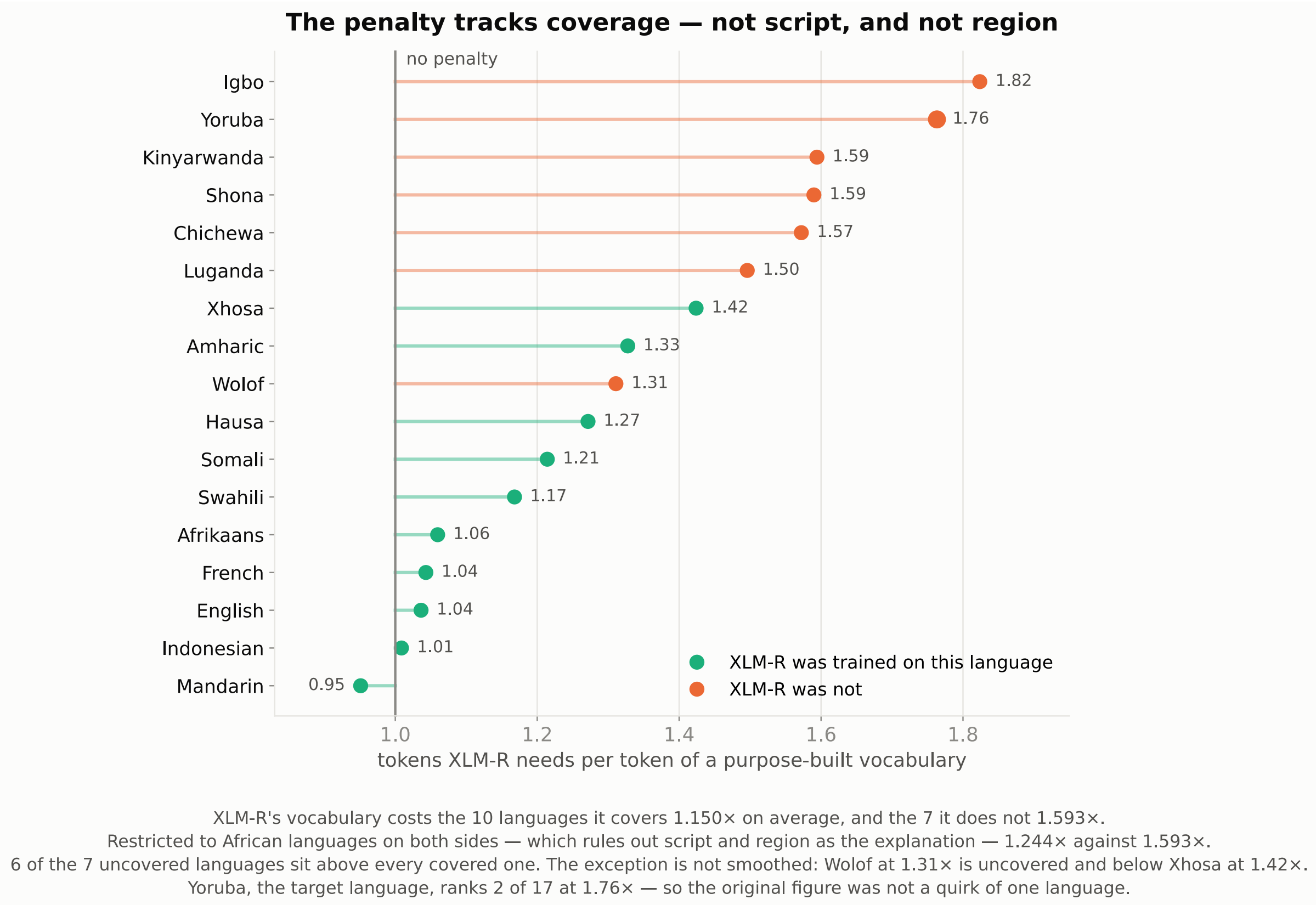
On SIB-200, the 33.8M from-scratch model reached 0.688 macro-F1 versus 0.582 for mmBERT (five seeds; learning rates chosen on the dev split). The 0.106 margin is much larger than seed noise, but bootstrap intervals overlap by 0.004: say “ahead,” not “beats.”

HEADLINE RESULT

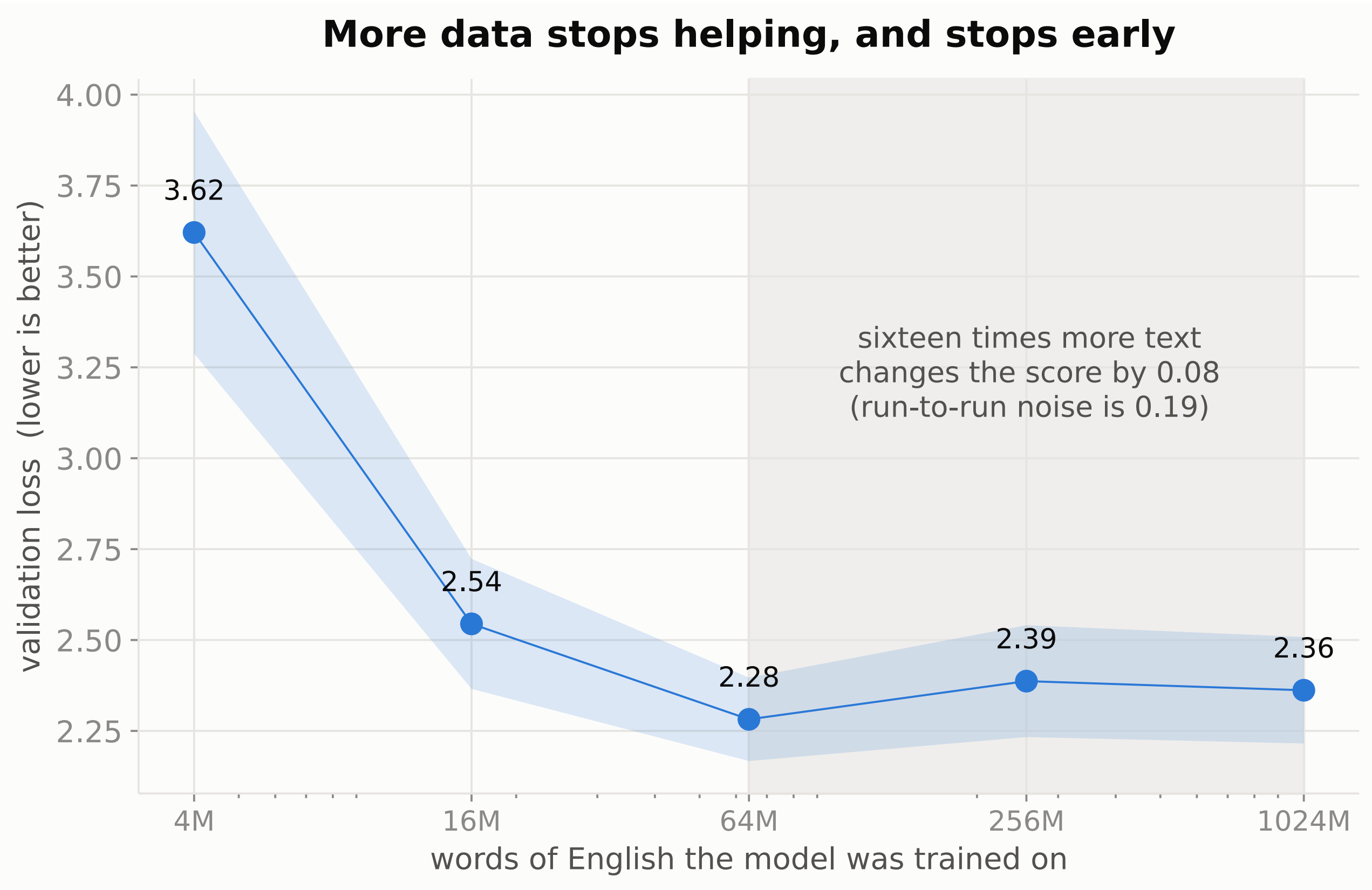
A 33.8M model trained only on Yoruba, against two much larger multilingual models



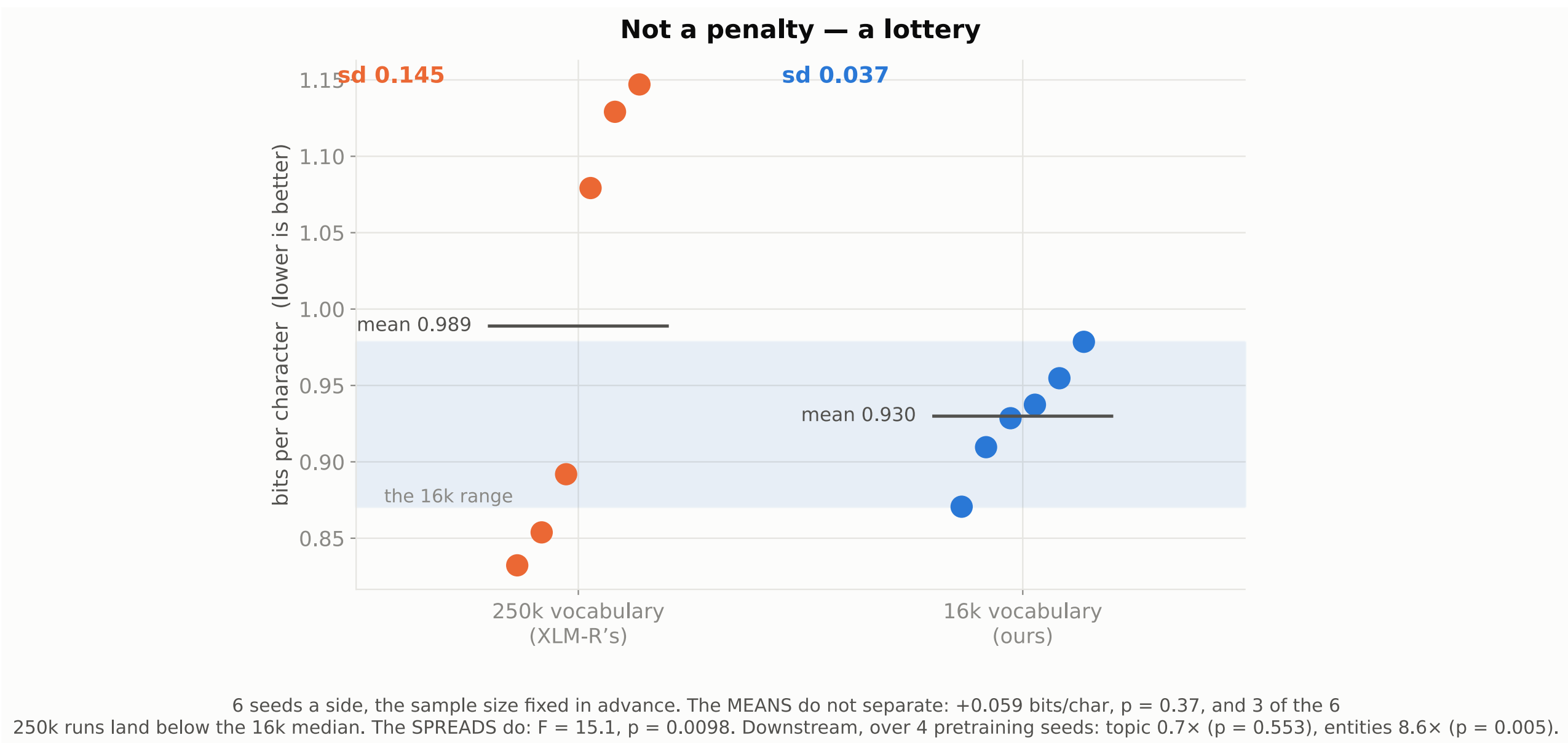
COVERAGE GRADIENT



DATA CONTROL



RUN-TO-RUN VARIABILITY



What survives

XLM-R needs 1.76 tokens for each token used by a Yoruba-specific 16k BPE—second-highest among 17 languages. A 128-token window holds about 44 Yoruba words with XLM-R versus 77 with the fitted vocabulary. Across languages, this penalty tracks XLM-R coverage.

Holding architecture, Yoruba text and compute fixed, swapping only the vocabulary changed downstream scores. The 16k vocabulary led the 250k XLM-R vocabulary by +0.144 on SIB-200 and +0.061 on MasakhaNER; every seed won in both tasks (four seeds per arm, exact $p=0.029$).

Boundary: Causal downstream evidence is still Yoruba-only. SIB-200 has just 204 test items, and NER did not receive the same dev-selection protocol. The next decisive test is the matched-compute vocabulary swap across four or five languages spanning the coverage gradient.